

NGUYEN Trong Thang, BEng



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ntthang-dev



My Blog



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NguyenTrongThang



Employment History

- 2023 – 2026 **Research Assistant**, Power System Laboratory, Ho Chi Minh City University of Technology (HCMUT).
- 2022 – 2024 **Product & Brand Consultant**, Private Enterprise, Ho Chi Minh City.
Focus: Product research and development (R&D), customer consultation, brand advertising, and marketing strategies.
- 2020 – 2022 **Research Assistant**, Power Electric Research Laboratory, Ho Chi Minh City University of Technology (HCMUT).
- 2021 **Engineering Intern**, Medium Voltage Consulting Department, HCMC Power Engineering Construction Consulting Joint Stock Company.

Education

- 2019 – 2026 **BEng.Electrical Engineering, Ho Chi Minh University of Technology** Specialization: Power Systems
GPA: 7.07/10.0 for 133/132 Credits | Credits earned: 191/132
– **Project 1:** Power Supply Design for a Brewery Factory - 9.0/10
– **Project 2:** Unified Multi-View Clustering Model for Time-Series Data: Integrating Shape Invariance and Outlier Robustness - 9.5/10
Thesis title: Proposing machine learning methods for classifying electricity customers with multiple characteristics - 9.7/10

Research Publications

Conference Proceedings

- 1 T. N. Trong, K. P. Nguyen, and M. Q. Huynh, "Information-theoretic multi-view clustering for robust and interpretable analysis of electrical load profiles," in *2025 International Symposium on Electrical and Electronics Engineering (ISEE)*, 2025, pp. 232–237. DOI: 10.1109/ISEE68370.2025.11223446.

Highlighted Software & Engineering Projects

- 2025 **Smart Grid Monitoring and Control System (Graduation Thesis)**
A comprehensive web platform for power system dispatch, electricity market simulation, and AI-driven customer load profiling.
 smart-grid-mivkm-thesis.netlify.app
- ClusterLayoutOptimizer (Electrical Cabinet Layout)**
Automated layout optimization system utilizing custom K-Means clustering and geometric algorithms for obstacle avoidance and automatic wiring.
 github.com/ntthang-dev/ClusterLayoutOptimizer

Highlighted Software & Engineering Projects (continued)

- 2024

Physics-Informed Neural Networks (PINNs) for Power System Stability

Pioneered the integration of physical laws into deep learning architectures to directly model and solve the complex dynamics of SMIB systems.
Technologies: Python, Deep Learning, Differential Equations.
- 2024

Small-Signal Stability via Matrix Decomposition

Formulated a mathematically rigorous, Big-O optimized computational tool applying matrix decomposition and eigen-analysis to block-diagonalize N-machine grid state matrices.
Technologies: Advanced Linear Algebra, MATLAB.
- 2024

Optimal Control of Automatic Voltage Regulator (AVR) via LQR

Designed a Linear Quadratic Regulator (LQR) algorithm to mathematically optimize the transient response and ensure robust voltage stability for synchronous generators.
Domain: Power System Stability, Optimal Control.
- 2024

Golang-based SCADA System via Modbus Protocol

Real-time data acquisition and remote control system for industrial electrical equipment using Golang and the Modbus communication protocol.
- 2025

Femnia: AI-Powered Women's Health Platform

An AI-integrated web application designed for tracking, predicting menstrual cycles, and providing personalized healthcare insights.
 femnia-v2.netlify.app
- 2024

Xiangqi (Chinese Chess) AI Engine & Web Platform

Intelligent game-playing AI utilizing Minimax, Alpha-Beta Pruning, and MCTS, fully deployed as an interactive web application.
 Core AI | Web App
- Probabilistic Robotics and 1D Robot Localization**

Implemented a 1D Particle Filter algorithm, utilizing Bayesian probability and sensor measurement updates to estimate the robot's state under extreme uncertainty.
Domain: Probability & Statistics, Autonomous Systems.
- Non-linear Inverted Pendulum Control via Sliding Mode Control (SMC)**

Developed a robust non-linear controller utilizing the Lyapunov stability criterion to maintain system equilibrium under high uncertainty and external disturbances.
Domain: Advanced Control Theory, Non-linear Dynamics.
- 2022

Convex Hulls in Normed Vector Spaces and Practical Applications

Conducted rigorous mathematical proofs on the geometric properties of convex hulls within infinite-dimensional normed spaces and explored their algorithm applications.
Domain: Functional Analysis, Convex Geometry.
- 2021

Vector and Matrix Norms: Theoretical Foundations and Applications in Lasso Regression

Investigated the mathematical properties of vector/matrix norms and their optimization role in L1-regularized (Lasso) machine learning models.
Domain: Advanced Linear Algebra, Machine Learning Optimization.
- 2020

The Standard Model of Particle Physics: Higgs Boson and Macroscopic Quantum States

Researched elementary particles, the Higgs mechanism, and the theoretical quantum physics underlying the 5th and 6th states of matter (Bose-Einstein and Fermionic condensates).
Domain: Quantum Physics, Particle Physics.
- Computational Astrodynamics: Multi-Stage Rocket Trajectory Simulation**

Programmed complex kinematic simulations of multi-stage rockets utilizing advanced numerical integration methods, including the Verlet algorithm and Iterative Euler.
Domain: Classical Mechanics, Numerical Analysis.

Skills & Expertise

Electrical Engineering	■ Power System Control & Stability, High Voltage & Lightning Protection, SCADA Systems & Modbus Protocol, Electrical Distribution, Power Supply System Design, IoT/Embedded, Arduino.
AI & Computer Science	■ Machine Learning (Regression, SVM, K-Means), Deep Learning (RNN, LSTM, MLP), TensorFlow, Reinforcement Learning (Q-Learning), Heuristic Search Algorithms (A*, MCTS), Data Structures & Algorithms.
Math, Physics & Quantum	■ Quantum Computing, Quantum Machine Learning, Advanced Linear Algebra, Functional Analysis, Complex Analysis, Wave Optics, Wave Physics, Nuclear Physics.
Programming Languages	■ C/C++, C#, Golang, Python, Rust, Java, JavaScript, R, Pascal, Fortran, x86/x64 Assembly, Basic, MATLAB, $\LaTeX$, PHP.
Web & App Dev	■ HTML/CSS, Flutter, Angular, XML/XSL, React, Node.js, Firebase, Figma.
Databases & DevOps / Tools	■ Linux, Git, Docker, MySQL, PostgreSQL, HSQL, SQLite.
Engineering Software	■ ETAP, AutoCAD, Autodesk Inventor, Altium Designer, Matlab Simulink, PSS/E, a little knowledge BIM MEP (Revit).
Professional Skills	■ Problem Solving, Academic Research, Project Management, Team Leadership, Teaching & Training, Consultation, Technical Writing & Publishing, $\LaTeX$ typesetting and publishing.
Languages	■ Vietnamese: Native English: Professional Working Proficiency. French: Conversational Basic Knowledge: Chinese, Korean, Japanese.

Miscellaneous Experience

Certification

- 2025 ■ **Semi-finalist, The 27th Euréka Scientific Research Student Award.**
Field: Economics (Commerce, Business Administration, and Marketing).
Project: "A Big Data Analytics Tool Utilizing Machine Learning for Energy Consumption Insight and Optimization in the Power Industry".
- **Certified of Completion the Global Project-Based Learning Program.** Awarded by Prof. Goro FUJITA Shibaura Institute of Technology, Japan.
- 2026 ■ **[Nominated] Best Graduation Thesis Award**, Faculty of Electrical and Electronics Engineering, HCMUT.
Recognized for achieving the highest evaluation score in the graduation defense committee (Pending final approval).

References

Dr. Khai Phuc Nguyen

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Ho Chi Minh City, Vietnam.
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